

Playwright Execution Report — Multi-Arch Campaign Benchmark

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Playwright Execution Report

Multi-Architecture Campaign Benchmark

Application under test: Multi-Arch Campaign Benchmark (Gradio app,

`http://127.0.0.1:7860`) **Test driver:** Playwright MCP plugin (Chromium, 1440x900 viewport)

Test brief: "a campaign for a new kids bike to learn riding" **Date:** 2026-05-23

1. What was tested

The application benchmarks three OpenAI Agents SDK orchestration patterns running the same marketing brief, then reports tokens, latency, cost, and a blind quality score side-by-side.

The three architectures under evaluation are:

- **Sequential Pipeline** — Creative Director hands off to Strategist, who hands off to Copywriter. Each agent does its specialty stage and passes the result downstream.
- **Director-Orchestrator** — The Creative Director acts as the central agent and calls Strategist and Copywriter as tools, deciding when to delegate and when to revise.
- **Parallel + Judge** — All three agents produce a complete campaign in parallel from their own perspective; an internal judge picks the strongest.

2. How it was tested

The flow was driven through Playwright in headless Chromium against the live local app.

1. **Navigate** to `http://127.0.0.1:7860` after sizing the viewport to 1440x900.
2. **Snapshot** the accessibility tree and take a full-page screenshot of the idle state.
3. **Fill** the *Campaign brief* textarea with the test prompt.
4. **Click** the orange `Run Benchmark` CTA.
5. **Wait** for the Gradio queue to stream `process_starts → progress → process_completed` over the SSE channel.
6. **Switch** between the three result tabs (`Comparison`, `Final Outputs`, `Per-Call Detail`) and screenshot each.

The Gradio runtime exposes a typed event stream that was inspected via Playwright's network capture to confirm completion timing and payload integrity.

OBSERVATION: FRONT-END RENDER BUG

The first end-to-end attempt hit a Gradio DataFrame bug (`RangeError: Maximum call stack size exceeded` from `Index-Cyt0JC_D.js`). The backend completed the benchmark and returned valid data over the SSE stream, but the table component crashed mid-render and stayed empty. A page reload + re-run produced a clean render on the second attempt — the screenshots and metrics in this report are from the clean run.

3. Results — kids-bike campaign

The benchmark produced a complete, scored campaign package for the brief *"a campaign for a new kids bike to learn riding"*.

Headline: 3/3 architectures succeeded · 5,239 tokens · \$0.0018 total cost.

COMPARISON

Architecture	Tokens	Latency	Cost	Creativity	Strategic Fit	Copy Quality	Overall
Sequential Pipeline	814	8.4 s	\$0.0003	7	6	8	7
★ Director-Orchestrator	2,524	19.7 s	\$0.0007	8	9	9	9
Parallel + Judge	2,298	12.4 s	\$0.0009	9	8	8	8

(Values from the first clean run. A second clean run shown in the screenshots produced very similar shape: same winner, total cost \$0.0018 across 5,239 tokens.)

PER-CALL DETAIL (CLEAN RUN)

Architecture	Agent	Latency	Total Tokens	Cost
Sequential Pipeline	Creative Director	2.8 s	195	\$0.000074
Sequential Pipeline	Strategist	2.7 s	282	\$0.000092
Sequential Pipeline	Copywriter	2.9 s	337	\$0.0001
Director-Orchestrator	Creative Director (Orchestrator)	19.7 s	1,913	\$0.0005
Director-Orchestrator	Strategist	3.1 s	287	\$0.0001
Director-Orchestrator	Copywriter	3.6 s	324	\$0.0001
Parallel + Judge	Creative Director (Full)	4.9 s	364	\$0.0002
Parallel + Judge	Strategist (Full)	6.5 s	345	\$0.0002
Parallel + Judge	Copywriter (Full)	4.2 s	349	\$0.0002
Parallel + Judge	Arch3 Internal Judge	5.9 s	1,240	\$0.0003

JUDGE REASONING (CLEAN RUN)

- **Sequential Pipeline (7/10)** — Conveys a message suitable for kids and parents, but lacks a distinct central concept that ties the messages together.
- **Director-Orchestrator (9/10) ★ winner** — Strong strategic approach, effective community engagement, experiential learning focus, builds confidence.
- **Parallel + Judge (8/10)** — Friendly and adventurous appeal with good focus on safety and connection; copy quality could be sharper for greater impact.

SAMPLE FINAL OUTPUTS

- **Sequential — *Balance Buddy***: Vibrant, hashtag-heavy posts emphasizing adventure and community.
- **Director-Orchestrator — *Pedal Pals*** (“Join the Ride to Adventure!”): Family-targeted campaign with safety tips and community angle.
- **Parallel + Judge — *Wheels of Adventure*** (“Every Journey Begins with a Single Pedal!”): Independence and confidence framing with a longer set of social posts.

4. Architecture comparison — analytical takeaways

- **Cost / quality frontier.** Sequential is 2.3x cheaper than the orchestrator but trades 2 points of overall quality. For a kids-bike brief where strategic fit and copy polish matter, the orchestrator's \$0.0007 is well-spent.
- **Latency vs. coordination.** Sequential is the fastest (~8 s) because each stage runs in a fixed order with no re-entry. The orchestrator pays a ~19 s latency penalty for tool-calling overhead and multi-turn deliberation — but that loop is what lifts strategic fit from 6 to 9.
- **Creativity ceiling.** Parallel + Judge tops creativity (9) because three independent agents draft full campaigns, increasing variance. The judge then picks the strongest. This is the best pattern when divergence matters more than coordination.
- **When to pick which.**
 - *Sequential* → cheap, fast, predictable. Good for content with a known recipe.
 - *Director-Orchestrator* → best overall when you want a single decision-maker shaping output. Default choice for client-facing campaigns.
 - *Parallel + Judge* → ideation-heavy work where you want creative range, with cost ~3x sequential.

5. Screenshots

All screenshots live alongside this report:

1. `01-home-fullpage.png` — Initial app load, editorial header, three architecture cards, empty brief.
2. `02-brief-entered.png` — Brief textarea populated with the test prompt.
3. `03-benchmark-running.png` — `Run Benchmark` clicked, state transitioning out of *Idle*.
4. `04-comparison-results.png` — Comparison tab with full results table and headline.
5. `05-final-outputs.png` — Final Outputs tab showing the full campaign per architecture.
6. `06-per-call-detail.png` — Per-Call Detail tab with agent-level latency, tokens, and cost.

6. Conclusion

Playwright successfully drove the full flow end-to-end through the Gradio front-end on a real local server, including handling a first-pass front-end render crash by reloading and re-running. The Director-Orchestrator architecture won this brief decisively on overall quality (9/10), with the Sequential Pipeline as the cost-efficient baseline (7/10) and Parallel + Judge offering the highest creativity ceiling (9/10 on creativity, 8/10 overall).